A small counting experiment

A walking robot leaves one marker at its first stop, two markers at its next stop, and one more at each later stop. We want a quick way to count the markers without drawing every stop.

Let n be the number of stops, and let $S(n)$ be the total marker count.
The following rule gives the answer.

$$S(n) = \frac{n(n + 1)}{2}$$

Read the product before dividing: pair the first short row with a reversed copy of the last long row. Every pair has the same length, so the rearranged picture forms a rectangle. Each original picture occupies half of that rectangle.

Check the idea with a loop

```
def triangular(stops):
    total = 0
    for step in range(1, stops + 1):
        total += step
    return total
```

Listing 1. A direct count used to check the rule.

The formula and the loop count the same arrangement in different ways. The formula avoids visiting every stop; the loop makes the counting process easy to inspect.

Try it: use four stops, and explain why two copies of the pattern always make a rectangle.

This page was written from scratch for a layout demonstration.